



ROBOCUPJUNIOR RESCUE LINE 2026

TEAM DESCRIPTION PAPER

Kavosh_Line

Abstract

Kavosh Rescue Line Robot is a compact, reliable, and competition-ready autonomous robot designed for accurate navigation, fast decision-making, and consistent rescue performance in the RoboCupJunior Rescue Line challenge. The robot is built around a Teensy 4.1 controller, a four-wheel DYNAMIXEL XC430 drive system, custom 3D-printed rubber-coated wheels, and a modular sensor layout. It uses 16 TCRT5000 reflective sensors for precise line tracking, three APDS9960 color sensors for floor-marker and evacuation-zone color detection, four VL53L1X time-of-flight distance sensors for obstacle and evacuation-zone awareness, and micro switches for reliable mechanical feedback. The main capabilities of the robot include stable line following, intersection decision-making, obstacle avoidance, ramp and gap handling, evacuation-zone entry, structured victim search, victim collection, and safe-zone placement. What sets the robot apart is the combination of mechanical simplicity, strong sensor coverage, fast processing, modular electronics, and easy repair during competition. Each subsystem was tested separately before full integration, which improved debugging and reliability under competition pressure. The project is also supported by Team Kavosh's long-term RoboCup experience, giving the team a strong background in building practical and robust autonomous rescue robots.

1. Introduction

a. Team

The team combines skills in mechanical design, electronics, programming, CAD modeling, and robotics competition experience. Ethan Chan contributes to mechanical development, Arduino-based programming, sensor integration, and Python testing/debugging, supported by four years of VEX Robotics experience. Huanyu "Carson" Gao focuses on mechanical design, assembly, electronics assembly, wiring support, and system integration, using tools such as Fusion 360, Blender, and SolidWorks. JayJay CJZ Zhou supports electronics and mechanical development through wiring, construction, module testing, and integration, with experience from FiraCup. By dividing responsibilities based on each member's strengths, the team can develop, test, and integrate the robot more effectively for the RoboCup Junior Rescue Line challenge.

2. Project Planning

a. Overall Project Plan

Our main objective for the RoboCup Junior Rescue Line competition is to build a compact, reliable rescue robot that can follow lines, detect intersections and markers, avoid obstacles, enter the evacuation zone, find victims, and complete the rescue task within the rules. The robot must be stable, easy to repair, and consistent under competition pressure. To achieve this, we set requirements for mechanics, electronics, and software. The robot needs a lightweight structure, stable wheels, well-placed sensors, safe wiring, and clear algorithms for line following, obstacle avoidance, evacuation search, victim detection, and recovery. Our plan followed milestones: build the structure, assemble and test electronics, develop software modules, and finally integrate everything through full-course testing.

Milestone	Description	Main Responsibility	Review Gate
Week 1–2: Requirement Analysis and Concept Design	Study the competition rules and define the robot’s dimensions, sensor requirements, rescue mechanism, and software functions.	Whole Team	Confirm that the proposed design satisfies all Rescue Line requirements.
Week 3–4: Mechanical Design and Assembly	Design and assemble the chassis, sensor mounts, motor placement, and victim-handling structure using CAD tools.	Whole Team	Check the robot’s stability, component accessibility, and ease of repair.
Week 5: Electronics Assembly	Connect the controller, sensors, motors, power system, and organize the wiring layout.	Carson and JayJay	Test each sensor and actuator individually before system integration.
Week 6–7: Software Development	Develop algorithms for line following, obstacle avoidance, marker detection, evacuation-zone searching, and victim handling.	Ethan	Test each software module separately using controlled scenarios.
Week 8: System Integration	Combine the mechanical, electronic, and software subsystems into one fully functioning robot.	Whole Team	Verify that all components communicate and operate correctly together.
Week 9–10: Reliability Testing and Improvement	Conduct repeated tests for line following, obstacle avoidance, evacuation-zone searching, and rescue tasks.	Whole Team	Record failures, identify weak modules, and improve the robot’s design and performance.

The team agreed on this schedule by analyzing the competition tasks and the dependencies between subsystems. For example, programming could not be fully finalized before the sensor placement and mechanical structure were stable. Each iteration was tested under specific conditions such as straight-line tracking, sharp turns, intersections, obstacle avoidance, victim detection, and evacuation zone sweeping. Results from each test affected the next design iteration. If a module failed repeatedly, the team reviewed whether the issue came from mechanical alignment, sensor position, wiring, or software logic before making changes.

b. Integration Plan

The robot was designed as one integrated system where mechanics, electronics, and software work together to complete the Rescue Line mission. The main controller reads data from line, color, distance, and victim detection sensors, then controls the motors and rescue mechanism based on the robot’s current task. Each component supports a specific competition requirement. Line sensors help with line following and gap recovery, color sensors detect markers and evacuation indicators, distance sensors detect obstacles and boundaries, and the motor system controls movement, turns, obstacle avoidance, and evacuation zone sweeping. During integration, each subsystem was tested separately before being combined step by step. This made debugging easier and helped identify hardware, wiring, calibration, or software problems. The final system supports stable movement, accurate tracking, reliable detection, correct decisions, obstacle handling, victim detection, and evacuation search.

3. Hardware

The mechanical design of our robot was developed to satisfy the main requirements of the RoboCup Junior Rescue Line competition: stable movement, accurate line following, reliable obstacle handling, and effective victim collection inside the evacuation zone. The robot was designed as a modular system so that the drivetrain, sensor mounts, electronics, and rescue mechanism can be built, tested, repaired, and improved separately before final integration.

a. Mechanical Design and Manufacturing

- **Main Structure:**

The main chassis forms the base of the robot and supports all major mechanical and electronic components. The structure was designed using CAD software such as Fusion 360 and SolidWorks before manufacturing. This allowed the team to check component placement, spacing, weight distribution, and accessibility before assembling the real robot. The chassis uses a layered and compact structure to keep the robot stable while still providing enough space for motors, sensors, wiring, the controller, and the rescue mechanism. The lower section holds the drivetrain and line-following sensor mounts close to the ground for accurate readings. The upper section supports the electronics and keeps the wiring organized and protected. This design improves reliability because important parts can be accessed quickly during testing or repairs.



Fig. 1 shows the general mechanical design of the robot in CAD and the assembled robot in the real world.

- **Actuators and Power Train:**

The robot uses a four-wheel drive system for better traction, stability, and control during line following, turns, ramps, gaps, and obstacle maneuvers. Each wheel has its own motor, allowing accurate forward movement and controlled rotations. The drivetrain focuses on repeatable motion and precise turning rather than only speed. Motor placement was adjusted to improve weight balance and reduce slipping during movement.

- **Wheels:**

The team designed and 3D-printed the wheels using PLA because it is lightweight and easy to modify. A rubber layer was added around each wheel to increase grip and reduce slipping during turns, ramps, and fast line-following corrections. The wheels were tested in straight motion, sharp turns, and start-stop movements. Based on testing, the surface and alignment were adjusted to improve traction and reduce drift.



Fig. 2 shows the robot wheel design.

- **Subassemblies and Modules:**

The robot is divided into several mechanical subassemblies:

- **Chassis Module:**

Holds the drivetrain, electronics, sensor mounts, and rescue mechanism in a compact structure.

- **Drivetrain Module:**
Includes the wheels, motors, and mounting points required for stable movement and accurate turning.
- **Rescue Mechanism Module:**
Includes the arm, gripper, guiding structure, and victim storage area used during evacuation zone tasks.
- **Sensor Mounting Module:**
Positions the line, color, and distance sensors at fixed heights and angles to improve measurement consistency

Dividing the robot into modules helped the team test and improve each subsystem independently. For example, the drivetrain could be tested before the rescue mechanism was finalized, and the sensor mounts could be adjusted without redesigning the full robot.

- **Rescue Mechanism**

The rescue mechanism was designed for detecting, guiding, and collecting victims in the evacuation zone. The mechanism uses servo-controlled arms and a gripper-style structure to approach and secure victims after detection. The front guiding parts help align the victim with the gripper, while the arm motion supports controlled pickup or movement into the storage area. The storage area was designed to keep collected victims inside the robot during transportation. Its shape was chosen to reduce the chance of victims falling out while the robot turns or moves across the evacuation zone. Cable management holes and protected mounting points were also included so that the mechanism can move without interfering with wires or sensors.

- **Design and Manufacturing Tools**

The team used CAD modeling to design and improve the robot before manufacturing. Fusion 360, SolidWorks, and Blender were used to create mechanical models, check clearances, visualize assemblies, and prepare parts for manufacturing. Several parts, including wheels, sensor holders, servo mounts, and rescue mechanism components, were 3D printed. This made the design process faster because the team could test a part, identify issues, redesign it, and print an improved version. This approach gave the team a competitive advantage because the robot was not built only from standard kit parts.

- **Mechanical Testing and Quality Assurance**

Mechanical testing was performed throughout the development process to validate the design and identify weak points before full-course testing. Each test focused on a specific requirement from the project plan.

Test	Purpose	Evaluation Method	Design Improvement
Chassis Stability Test	Check whether the robot remains balanced during movement and turning.	Drive the robot forward, stop it quickly, and rotate it in place.	Adjusted component placement and improved weight distribution.
Wheel Traction Test	Reduce wheel slipping on the competition surface.	Test straight-line motion, turning, and movement on ramps.	Added a rubber layer to the wheel surfaces to increase traction.
Turning Accuracy Test	Improve maneuvering at intersections and around obstacles.	Perform repeated 90-degree and 180-degree turns.	Adjusted wheel alignment and refined motor mounting.
Sensor Height Test	Improve the consistency of line and color detection.	Compare sensor readings at different mounting heights.	Redesigned the sensor mounts to provide more stable readings.
Rescue Mechanism Test	Verify the reliability of victim pickup, storage, and release.	Perform repeated victim-approach, pickup, storage, and release trials.	Improved the guiding structure and gripper alignment.
Durability Test	Evaluate the robot's reliability after repeated operation.	Complete multiple practice runs and inspect the robot after each run.	Reinforced component mounts and improved cable routing.

The results of these tests directly affected later design iterations. For example, when wheel slipping reduced turning accuracy, the wheel surface was improved with a rubber layer. When sensor readings changed because of vibration or height variation, the sensor mounts were redesigned to hold the sensors more rigidly. When the rescue mechanism had alignment problems, the guiding structure and gripper position were adjusted to make victim collection more repeatable. Overall, the mechanical design was developed through an iterative process of CAD design, manufacturing, testing, and improvement. The final structure is modular, repairable, and suitable for the main Rescue Line challenges: navigation, stable motion, obstacle handling, evacuation zone search, and victim rescue.

- **Motors:**

Our robot uses DYNAMIXEL smart servomotors for both locomotion and the rescue mechanism. Four DYNAMIXEL XC430 motors are used for the four-wheel drive system. Each wheel has its own motor, which allows accurate speed control, repeatable turns, and encoder-based movement correction during line following, intersections, obstacle avoidance, and evacuation-zone sweeping. One DYNAMIXEL AX-12A is used as the main rescue arm actuator because it provides enough torque for controlled lifting and positioning of the mechanism. Three DYNAMIXEL XL-320 servomotors are used for the lighter rescue mechanism functions: two motors control the gripper fingers and one motor controls the victim basket/storage mechanism. Using DYNAMIXEL actuators across the robot simplifies communication, debugging, and position control because the motors can be commanded through the same serial control structure. The actuator system was tested by running repeated straight-line movements, 90-degree turns, 180-degree turns, gripper open/close cycles, arm movement cycles, and basket release cycles. These tests helped the team tune motor speed limits, check mechanical alignment, reduce wheel slip, and confirm that the rescue mechanism could move without interfering with sensors, wiring, or the chassis.

b. Electronic Design and Manufacturing

The electronic system of our robot was designed to provide reliable sensing, stable power distribution, organized communication, and accurate actuator control for the RoboCup Junior Rescue Line challenge. The main goal of the electronic design is to support line following, marker detection, obstacle detection, victim detection, evacuation zone search, and rescue mechanism control. To improve reliability and debugging, the electronics were divided into several functional modules: the main controller, line sensor board, color detection sensors, distance sensing system, contact sensors, power regulation system, and actuator interfaces.

- **Main Controller:**

The robot uses a Teensy 4.1 microcontroller as the main processing unit. It reads data from the sensors, analyzes the robot's current state, and sends commands to the motors and rescue mechanism. The Teensy 4.1 runs the main control algorithms for line following, intersection handling, obstacle avoidance, evacuation zone search, victim detection, and rescue actions. The Teensy 4.1 was selected because it provides high processing speed, many input/output pins, and reliable communication options for connecting multiple sensors and actuators. Its fast performance helps the robot react quickly during line tracking, sharp turns, and rescue tasks. The software is written mainly in Arduino/C++ style, which allows each sensor and module to be tested separately before full-system integration.

- **Sensors and Modules**

The robot uses multiple sensor types because each part of the Rescue Line challenge requires different environmental information. The sensor system is divided into four main groups: line tracking, color detection, distance sensing, and mechanical feedback.

- **Push Buttons and Micro Switches**

The robot uses different sensors for different Rescue Line tasks. Four micro switches are used as contact sensors to detect physical **contact** or mechanism position. They provide simple ON/OFF feedback, making them reliable and easy to debug.

- Line Tracking Sensor:**
 For line tracking, the robot uses 16 TCRT5000 infrared reflective sensors. These sensors detect the contrast between the black line and the white floor. The main sensor array is positioned close to the ground to improve reading stability, while additional sensors help with sharp turns, intersections, line gaps, and recovery situations. During testing, the team adjusted the sensor height, angle, and software thresholds to reduce false readings caused by lighting changes, surface reflection, and vibration. The software uses the sensor readings to estimate the robot's position relative to the line and correct the left and right motor speeds.
- Color Detection Sensors**
 The robot uses three APDS9960 color sensors. Two APDS9960 sensors are mounted underneath the robot to detect floor markers such as green and red markers used for navigation decisions. The third APDS9960 sensor is mounted near the front/top area of the robot to detect the color of the evacuation safe zone, which can be red or green depending on the rescue task. The APDS9960 sensors were selected because they are compact, digital, and suitable for detecting RGB color information. During testing, the team calibrated color thresholds under different lighting conditions to improve marker recognition and reduce false detections.
- Distance Sensing System**
 For distance sensing, the robot uses four VL53L1X time-of-flight sensors. Two VL53L1X sensors are mounted at the front of the robot to detect obstacles and help the robot stop before collision. The other two VL53L1X sensors are mounted on the sides to support wall, boundary, and victim-search awareness inside the evacuation zone. These sensors provide fast and compact distance measurements, making them suitable for obstacle avoidance and structured sweeping in the evacuation area. The distance thresholds were tested and tuned so that the robot can detect objects early enough while avoiding false triggers from the floor, walls, or angled surfaces.
- Main Controller and Communication**
 The main controller of the robot is a Teensy 4.1, which acts as the central processing unit of the electronic system. It was selected because it provides high processing speed, many input/output pins, and reliable communication options for handling multiple sensors and actuators in real time. The Teensy 4.1 receives analog readings from the 16 TCRT5000 line sensors, digital color information from the three APDS9960 sensors, distance measurements from the four VL53L1X sensors, and digital feedback from the micro switches. Based on these inputs, the controller runs the robot's state-machine algorithm and sends commands to the four XC430 drive motors, the AX-12A rescue arm motor, and the three XL-320 motors used for the gripper and basket mechanism. This communication structure allows the robot to connect sensing, decision-making, movement, and rescue actions in one integrated system.

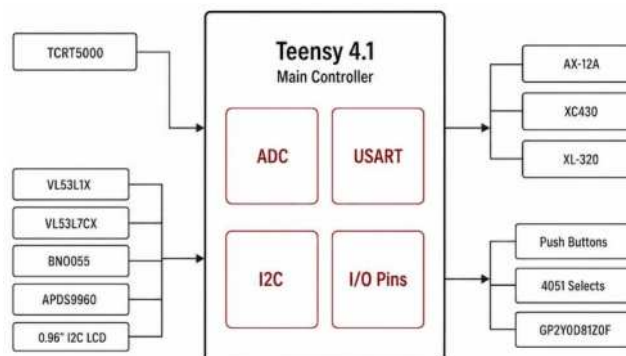


Fig. 3: Main controller communications

- **Power Subsystem**

The robot is powered by an 11.1 V, 3-cell lithium-polymer battery. Since different components require different voltages, the power system is divided into regulated and direct supply rails. The DYNAMIXEL motors are powered from the main battery supply according to their operating requirements, while the sensors and controller are powered through regulated voltage rails. A 5 V rail is generated for 5 V components, and a separate 3.3 V rail is generated using a Pololu D24V22F3 step-down regulator for 3.3 V sensors and logic-level devices. Separating the sensor power from the high-current motor supply improves reliability and reduces electrical noise. The wiring layout was organized so that power lines, signal lines, and actuator cables are easier to inspect and repair during testing and competition.

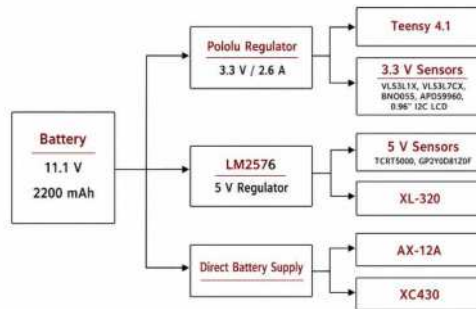


Fig. 4: Power Diagram

- **Circuit and PCB Design**

The electronic circuits were designed using Altium Designer. The PCB includes organized connectors, sensor connections, power regulation, and controller interfaces. This custom circuit design reduces loose wiring and makes the robot easier to assemble, test, and repair during competition. The modular electronic layout also supports better integration with the mechanical design, because sensor positions, connector locations, and board mounting points were planned together with the robot structure.

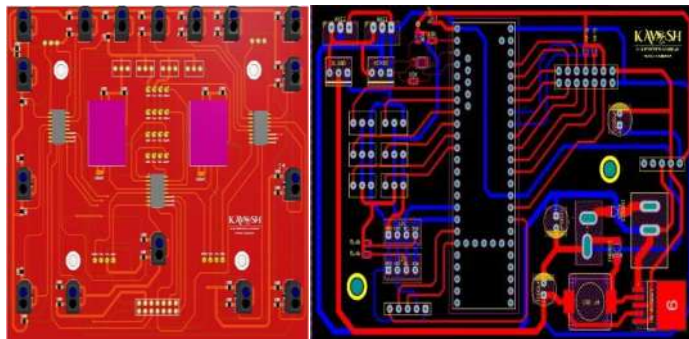


Fig. 5: Sensor board (3D) and Main board (2D) in Altium designer

- **Testing:**

Our testing procedures are precise and systematic, ensuring the reliability and functionality of our electronic design. Each component and subsystem faces comprehensive testing to validate performance and identify potential issues.

- **Component Testing:** Individual components, such as sensors and microcontrollers, are subjected to functional tests to verify proper operation and compatibility with other elements of the system.
- **Subsystem Integration:** Subsystems, including power sources and circuitry, are integrated and tested to ensure seamless interaction and compatibility within the overall system architecture.
- **Functional Testing:** The entire electronic system is tested under various operating conditions to evaluate its functionality, responsiveness, and accuracy. This involves simulating real-world scenarios to assess performance under different environmental factors.

- **Reliability Testing:** Extensive reliability tests are conducted to assess the durability and robustness of the electronic design. This includes stress testing under extreme conditions to identify potential failure points and weaknesses.
- **Innovative Solutions:**
Our electronic design incorporates innovative solutions and approaches to address specific challenges and optimize performance.
 - **Advanced Power Management:**
Innovative power management techniques are employed to optimize energy efficiency and prolong battery life, ensuring prolonged operation in demanding environments.
 - **Modular Design Philosophy:**
A modular design philosophy is adopted to facilitate scalability and flexibility, allowing for seamless integration of additional features and functionalities as needed.

These innovative solutions and approaches demonstrate our commitment to pushing the boundaries of electronic design and manufacturing, resulting in a robust and reliable system capable of meeting the demands of our project requirements.

4. Software

a. General software architecture

The software of our robot was designed with a modular state-machine structure. Each major competition task is handled by a separate software module, and the main program switches between these modules based on sensor readings and the current mission state. This structure makes the code easier to test, debug, and improve without changing the entire program. The main software modules are:

- **Line Following:**

The robot uses 16 TCRT5000 infrared sensors to detect the black line and estimate its position relative to the center of the robot. Each sensor is read and compared with a calibrated threshold. The software calculates the line position using a weighted sensor method, where sensors closer to the left or right side create a larger correction value. This error value is then used to adjust the speed difference between the left and right XC430 drive motors. When the line is centered, both sides move at nearly the same speed. When the line shifts left or right, the robot changes the motor speeds to return to the center.

- **Marker Detection:**

Three APDS9960 color sensors are used for detecting colored field markers. Two bottom sensors detect green and red markers on the floor for intersection decisions. Another APDS9960 sensor is used in the evacuation zone to detect the green safe area where the victim must be placed. The color thresholds were calibrated through repeated tests under different lighting conditions.

- **Obstacle Handling:**

The robot uses VL53L1X time-of-flight distance sensors to detect obstacles and walls. When an obstacle is detected, the robot stops, adjusts its position, avoids the obstacle, and then searches for the line again. This allows the robot to continue the mission without losing the path.

- **Orientation and Recovery:**

The BNO055 orientation sensor is used to improve turning accuracy and recovery movements. It helps the robot keep track of its heading during turns, obstacle avoidance, and evacuation zone movement, especially when the line is temporarily unavailable.

- **Evacuation Zone Search:**

When the robot enters the evacuation zone, it switches from line-following mode to search mode. The robot uses a sweeping method to cover the area in structured lanes. It moves across the zone, shifts sideways by one sweep width, reverses direction, and continues until the search area is covered. This method reduces random movement and increases the chance of finding victims efficiently.

- **Victim Detection and Collection:**

The VL53L7CX matrix distance sensor is used to detect object position and shape inside the evacuation zone. Since it provides multiple distance measurements, the robot can better estimate where a victim is located.

- **Integration:**

All modules are connected through the main state machine running on Teensy 4.1. This modular architecture allows the robot to switch between line following, marker detection, obstacle avoidance, evacuation zone sweeping, victim collection, and safe-zone placement.

b. Innovative solutions

One of the main software innovations is the use of a structured sweeping method for evacuation zone search. Instead of moving randomly, the robot covers the zone lane by lane. This makes the search more predictable, reduces repeated coverage of the same area, and improves the chance of detecting victims. Another important solution is the use of matrix-based distance data from the VL53L7CX sensor. The robot also uses multi-sensor decision-making. Line sensors, color sensors, distance sensors, orientation feedback, contact switches, and gripper confirmation sensing are combined to reduce false decisions. For example, the rescue mechanism is only activated when the software receives a valid victim detection and confirmation from the gripper sensor. The software was tested in stages. First, each module was tested separately, such as line following, color detection, distance sensing, and victim confirmation. Then, the modules were integrated step by step. Full-course testing was used to check whether the robot could switch correctly between states during realistic competition conditions.

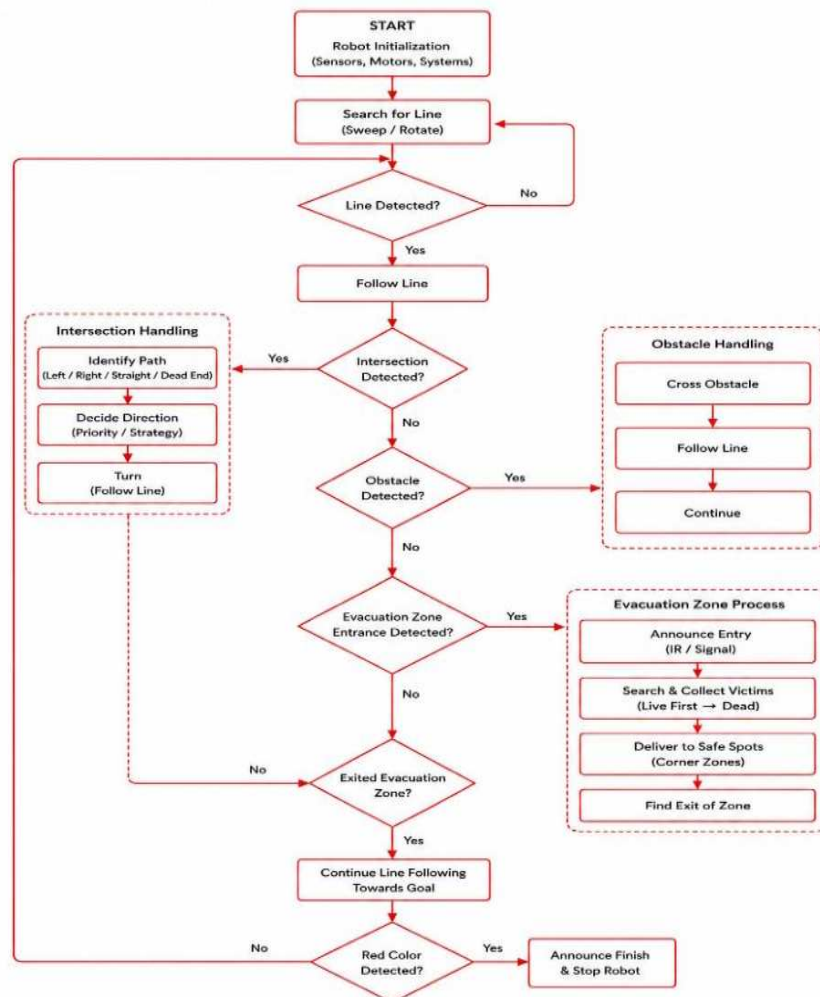


Fig. 6: The flowchart of how robot operates

5. Performance evaluation

The robot's performance was evaluated against the main RoboCupJunior Rescue Line challenges: line following, marker detection, obstacle handling, evacuation-zone search, victim collection, and rescue mechanism operation. Testing was performed in repeated stages so the team could identify weak modules and improve both hardware and software before full integration.

Test	Purpose	Evaluation Method	Design Improvement
Line following test	Check stability and path accuracy	Repeated straight lines, curves, and sharp turns	TCRT5000 thresholds and sensor height were adjusted to improve line stability
Intersection test	Verify marker-based decisions	Test left, right, and dead-end markers using bottom APDS9960 sensors	Color thresholds were calibrated under different lighting conditions
Obstacle avoidance test	Confirm safe obstacle bypass	Place obstacles in front of the robot and test recovery to the line	VL53L1X distance thresholds and recovery turns were tuned
Gap recovery test	Check line-loss recovery	Remove short sections of line and test re-detection behavior	The recovery search routine was improved using last-known line direction
Evacuation-zone search test	Evaluate structured victim search	Test sweeping movement inside a simulated evacuation zone	Side VL53L1X readings helped keep the search pattern more organized
Rescue mechanism test	Verify victim pickup and storage	Run repeated arm, gripper, and basket motion cycles	Gripper alignment and basket timing were adjusted for more reliable collection
Full integration test	Check complete mission behavior	Run line following, obstacle avoidance, evacuation search, and rescue actions together	Hardware calibration and software state transitions were refined

The testing process showed that the robot performed best when each subsystem was calibrated separately before full-course testing. For example, unstable line following was improved by adjusting TCRT5000 thresholds and sensor height. False color detections were reduced by calibrating APDS9960 values under different lighting conditions. Obstacle avoidance became more reliable after tuning the VL53L1X trigger distance and recovery movement. In the evacuation zone, the structured sweeping method made the robot's movement more organized and easier to debug compared to random search. These results showed that the robot's reliability depends on both mechanical alignment and software calibration, so the team recorded failures after each test and used them to guide the next design iteration.

6. Conclusion

In conclusion, this project focused on designing and building a reliable RoboCupJunior Rescue Line robot capable of line following, marker detection, obstacle handling, evacuation-zone search, victim collection, and rescue mechanism control. The final robot combines a modular mechanical structure, a Teensy 4.1-based electronic system, DYNAMIXEL smart actuators, and a state-machine-based software architecture. The use of 16 TCRT5000 sensors, three APDS9960 color sensors, four VL53L1X distance sensors, XC430 drive motors, and a servo-controlled rescue mechanism gives the robot strong sensing and control capability while keeping the design repairable and practical for competition. A key strength of the robot is its organized evacuation-zone strategy, where structured sweeping and distance feedback are used instead of random movement. The project improved the team's skills in CAD design, electronics, programming, sensor calibration, mechanical testing, and full-system integration. Future improvements will focus on increasing search speed, improving victim collection consistency, refining color calibration under different lighting conditions, and performing more full-course testing. Overall, the project demonstrates how careful planning, modular design, and continuous testing can produce a stronger and more competition-ready rescue robot.

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